



Research article

Facile and green synthesis of recyclable, environmentally friendly, chemically stable, and cost-effective magnetic nanohybrid adsorbent for tetracycline adsorption



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ABSTRACT

Antibiotic contamination of water sources, particularly tetracycline (TC) contamination, has emerged as one of the global issues that needs action. In this research, $\text{ZnCoFe}_2\text{O}_4@\text{Chitosan}$ (Ch) as a magnetic nanohybrid adsorbent was synthesized using the microwave-assisted co-precipitation method, and their efficiency for the TC adsorption process was investigated. FESEM (Field Emission Scanning Electron Microscope), EDX (Energy Dispersive X-ray), Mapping and line Scan, XRD (X-Ray Diffraction), FTIR (Fourier Transform Infrared Spectrometer), VSM (Vibrating Sample Magnetometer), Thermogravimetric analysis (TGA) and BET (Brunauer Emmett Teller) techniques were used to check and verify its physical and chemical properties. The removal of TC via the adsorption process from synthetic and real wastewater samples was investigated. The factors determining the TC adsorption process, comprising tetracycline concentration (5–30 mg/L), adsorbent dosage (0.7–2 g/L), contact time (2–45 min), and pH (3–11), were evaluated. The removal effectiveness for the synthetic sample and the real wastewater sample was 93 % and 80 %, respectively, under the ideal TC adsorption process parameters of pH 3, adsorbent dosage 1 g/L, TC initial concentration 5 mg/L, and contact time 30 min. According to kinetic and equilibrium studies, the adsorption of TC by $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ follows *pseudo*-second-order kinetics and the Freundlich isotherm. Additionally, it was determined through the analysis of thermodynamic data that the process of exothermic adsorption is spontaneous and is followed by a decrease in disorder ($\Delta H = -15.16$ kJ/mol, $\Delta S = -28.69$ kJ/mol, and $\Delta G = -6.62$ kJ/mol). After five cycles of recovery and regeneration, the $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nanocomposite was able to remove 65 % of the TC pollutant and had good chemical stability. The results showed that the magnetic nano-adsorbent $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ is a novel magnetic nano-adsorbent with high adsorption capacity that can be utilized to eliminate pharmaceutical contaminants from aqueous solutions.

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1. Introduction

The discharge of industrial effluents comprising organic and inorganic chemicals into environments is one of the foremost noteworthy sources of contamination. Pharmaceutical products are among the persistent pollutants in wastewater, and the prevalence of various pharmaceutical product subgroups in aquatic environments has drawn more attention [1–5]. In common, there are numerous diverse ways that pharmaceutical products are presented into the environment, including rural runoff, coordinated release from civil and healing center wastewater treatment facilities, coordinated release of therapeutic, veterinary, and industrial waste, and dumping of expired pharmaceutical products [6]. Low concentrations of antibiotics will cause antibiotic resistance and in high concentrations, the compounds are toxic to microorganisms. Although these substances only exist in extremely small amounts in water nanograms or micrograms per liter—their accumulation in plants, animals, and humans can eventually harm live organisms [7]. Antibiotics are a type of medicine that is frequently used in human societies to both prevent and cure bacterial infections. Some of these substances' metabolites are eliminated from the body and discharged into the environment because the human body cannot fully metabolize them. This increases the requirement to recognize and eliminate these substances from aquatic habitats and water supplies [6].

Tetracyclines (TC) are broad-spectrum antibiotics with $C_{22}H_{24}N_2O_8$ molecular formula used to treat gram-positive and gram-negative bacteria as well as intracellular chlamydia, mycoplasma, and rickettsia and it is regarded as the second-most utilized antibiotic in the world due to benefits including lower prices and better antibacterial activity [8]. Tetracycline concentrations are estimated to be 15 $\mu\text{g/L}$ in surface and groundwater, 86–199 μg in soil, 4 $\mu\text{g/L}$ in liquid fertilizers, and 3 $\mu\text{g/L}$ in wetlands [9]. On the other hand, antibiotics can produce undesirable byproducts when water is chlorinated and ozonated in a wastewater treatment facility [10]. Therefore, it is necessary to purify water and wastewater that contain TC antibiotic before releasing them into the environment.

The removal of antibiotics from wastewater can be accomplished in a variety of methods, including physical, chemical, and biological [11,12]. The method chosen will depend on the chemical and physical properties of the material, the required efficiency, and the planned use of the treated effluent [13,14]. These techniques consist of ultraviolet photolysis [15], surface adsorption [16–18], biodegradation [19], photocatalytic degradation [20,21], advanced oxidation processes during Fenton processes [22], photo-Fenton [23], electro-Fenton [24], photo-electro-Fenton [25], ozonation [26], immobilization of catalytic degrading enzyme [27], and catalytic ozonation [28]. Despite the benefits, each of these techniques also has drawbacks that frequently make it challenging to employ them.

The adsorption process is a promising method for reducing pollutants due to its low operating costs, high flexibility, non-sensitivity to toxic compounds and contaminants, lack of secondary pollution in the system, possibility of adsorbent separation, effectiveness, and high specific surface area for removal of pharmaceutical products [29], dye compounds [30], and heavy metals [31]. The majority of solid porous adsorbents used in adsorption operations have pores that are in the nanoscale range [32]. Nanomaterials' large surface area and the presence of active sites give them a great capacity for adsorption [33]. Additionally, magnetic nanosorbents can be separated and reused afterward [34].

The development and synthesis of magnetic adsorbents, iron oxides, and other ferrite compounds have drawn a lot of interest [35–39]. Due to their excellent adsorption effectiveness, cheap production cost, quick separation, low pollutant generation, and higher treatment efficiency in a short amount of time, magnetic adsorbents have found a good role in the treatment of water and wastewater [40]. MOF@GO [41], SF- Fe_3O_4 -EDTA [42], CoFe_2O_4 @ γ - Fe_2O_3 [43], Thioglycolic-modified Zn-Fe layer double hydroxide [44], CoFe_2O_4 @ SiO_2 -C₈ [45], GO/g-C₃N₄- Fe_3O_4 [46], Fe_3O_4 @ SiO_2 @CS/GO [47], Ni/HAP/ CoFe_2O_4 [48], and AC- Fe_3O_4 [49] are different magnetic nanosorbents that have been applied thus far.

In order to enhance the structure and function of these magnetic nanosorbents, biological substances can be used in their synthesis [36,37,50]. Chitin is used to produce the natural amino polysaccharide known as chitosan (Ch) [51,52]. Following cellulose, chitin is the second most prevalent polymer in nature. It may be taken from the shells of many crustaceans, including insects, fungi, and shrimp. Due to its unique qualities, including hydrophilicity, biocompatibility, biodegradability, non-toxicity, and most crucially the characteristics of high adsorption in the synthesis of nanosorbents, chitosan should be employed as a biopolymer [53].

Cellulose-derived carbohydrate biopolymers have a wide range of uses in water and wastewater treatment procedures because they are biodegradable, affordable, readily accessible, regenerative, stable, non-toxic, and have a high permeability. By magnetizing these polymeric carbohydrates and synthesizing ferrite nano-biocomposites, lumpy and low surface area issues can be resolved.

In the adsorbent structure, metal ions of transition metals including cobalt, zinc, and iron can function as either Lewis's acids or electron receptors. Functional groups in the TC structure like hydroxyl, amine, and carbonyl may also act as electron donor groups (EDGs) for transition metals such as cobalt, zinc, and iron. The EDGs of TC interact with the metal ions of the transition metals as electron-withdrawing groups (EWGs) to form a complex between the TC and the adsorbent. This complex's formation could affect the TC adsorption procedure [54,55].

In this research, a new magnetic material called $\text{ZnCoFe}_2\text{O}_4$ @Ch was made using a biopolymer called Ch. It can be used to remove TC from water. Adding Ch during the making of $\text{ZnCoFe}_2\text{O}_4$ @Ch makes the magnetic material have a bigger surface area. Furthermore, the amine and hydroxyl groups in Ch create a strong pull towards the adsorbent surface. This means that more TC molecules stick to the $\text{ZnCoFe}_2\text{O}_4$ @Ch surface and interact with it, which helps the adsorption process. Also, we looked at how different factors like the acidity of the solution, the amount of TC in the beginning, the amount of adsorbent used, and how long they were in contact, affected how well the adsorbent worked. We also studied how quickly the adsorbent worked and the energy involved in the process. The reusability and chemical stability of nano-biocomposite were also assessed.

2. Materials and methods

2.1. Chemicals

Cobalt (II) chloride hexahydrate ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$), iron (III) chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$), zinc chloride (ZnCl_2), sodium hydroxide (NaOH), chitosan (Ch) and hydrogen chloride (HCl) were purchased by Sigma Aldrich and Merck companies. The tetracycline was provided by Darou Pakhsh Pharmaceutical Manufacturing Company (Tehran, IRAN). Deionized water was used to make the experiment solutions. HCl and NaOH (1 N) were used to alter the pH of the solutions.

2.2. Characterizations

A microwave oven (SAMSUNG, 2450 MHz, 800 W) was applied to synthesize the magnetic nano adsorbent. XRD to determine the magnetic nano-crystal adsorbent's structure and phase (PHILIPS PW1730), FESEM to investigate the structure, composition, and surface structure of magnetic nano-adsorbents on the nanoscale (TESCAN MIRA III), EDX, Mapping and Line scan to investigate the element weight percentages, types, and distributions at the surface of magnetic nano-adsorbent (TESCAN MIRA II, SAMX Detector), FTIR to investigate chemical bonding in magnetic nano-adsorbents and determine functional groupings (AVATAR, Thermo), VSM to determine the magnetic properties of magnetic nano-adsorbent (LBKFB, Kashan Kavir Magnet Company), TGA (SDT, Q600) to determine the adsorbent's thermal stability, and BET to determine the size of a magnetic nano adsorbents specific region (BELSORP MINI II), techniques were used to study the structure of the resultant adsorbent. After confirming the magnetic nano adsorbent's physical and chemical structure, it was used to adsorb TC from an aqueous medium. A UV-vis spectrophotometer (SHIMADZU, UV-1800) was applied to detect the concentration of TC at a maximum wavelength of 357 nm ($\lambda_{\text{max}} = 357 \text{ nm}$).

2.3. Preparation of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$

$\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ nanocomposite was synthesized by modifying previously reported methods [56,57]. Stoichiometric amounts of metal chlorides (5.4 g $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, 1.2 g $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, and 0.7 g ZnCl_2) and chitosan (1 g) were respectively dissolved in 100 mL of deionized water. Then 6 g of NaOH was gradually added to the mixture over the course of an hour to alkalize the pH of the solution. Then the resulting mixture was placed in the microwave at 450-W 3 times for 5 min to obtain a black precipitate. The resulting sediment was separated using a magnet and rinsed several times with deionized water to neutralize the final pH. Eventually, the black precipitate was dried for 24 h at 70 °C temperature (in the furnace).

2.4. Batch adsorption experiments

At first, TC stock solution was prepared with a concentration of 100 mg/L and then concentrations of 10, 15, 20, 30, and 40 mg/L were prepared from it. To optimize the adsorbent dose, amounts of (0.7, 1, 1.5, 2) g/L were investigated. Solutions of 1 N NaOH and HCl were used to adjust the pH and pHs of 3, 5, 7, 9, and 11 were investigated. Contact times of 45 min and temperatures of 25, 35, 45, and 55 °C were investigated and optimized. Fig. 1 shows the batch adsorption setup schematic. All the experiments were performed on the synthetic samples. $\text{ZnCoFe}_2\text{O}_4$ alone and Ch alone were utilized as the controls to assess the superior performance of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nanocomposite. After optimizing the adsorption process conditions on the synthetic samples, the TC adsorption

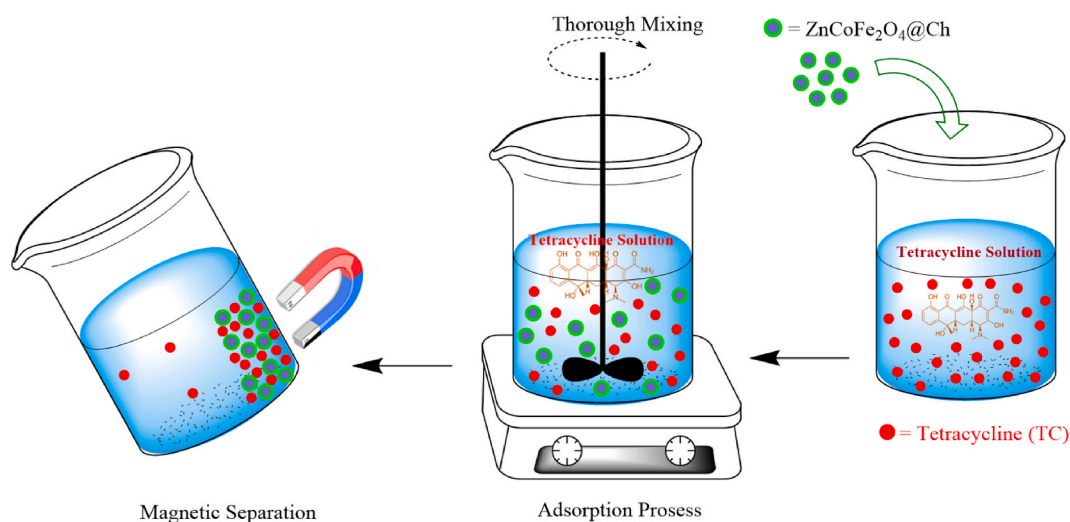


Fig. 1. Batch adsorption process setup schematic.

process has been carried out under optimal conditions on the Kerman University of Medical Sciences (Kerman, IRAN) with coordinates 30°.24' latitude and 57°.10' longitude wastewater sample as a real sample.

Finally, the adsorption efficiency (Eq. (1)) and adsorption capacity (Eq. (2)) were calculated by the following equations [58]:

$$R_e (\%) = \frac{C_0 - C_t}{C_0} \times 100 \quad (\text{Eq. 1})$$

R_e : adsorption efficiency (%)

C_0 and C_t : initial concentration and final concentration (mg/L) of TC, respectively.

$$Q_e = \frac{(C_0 - C_t)V}{m} \quad (\text{Eq. 2})$$

Q_e : adsorption capacity

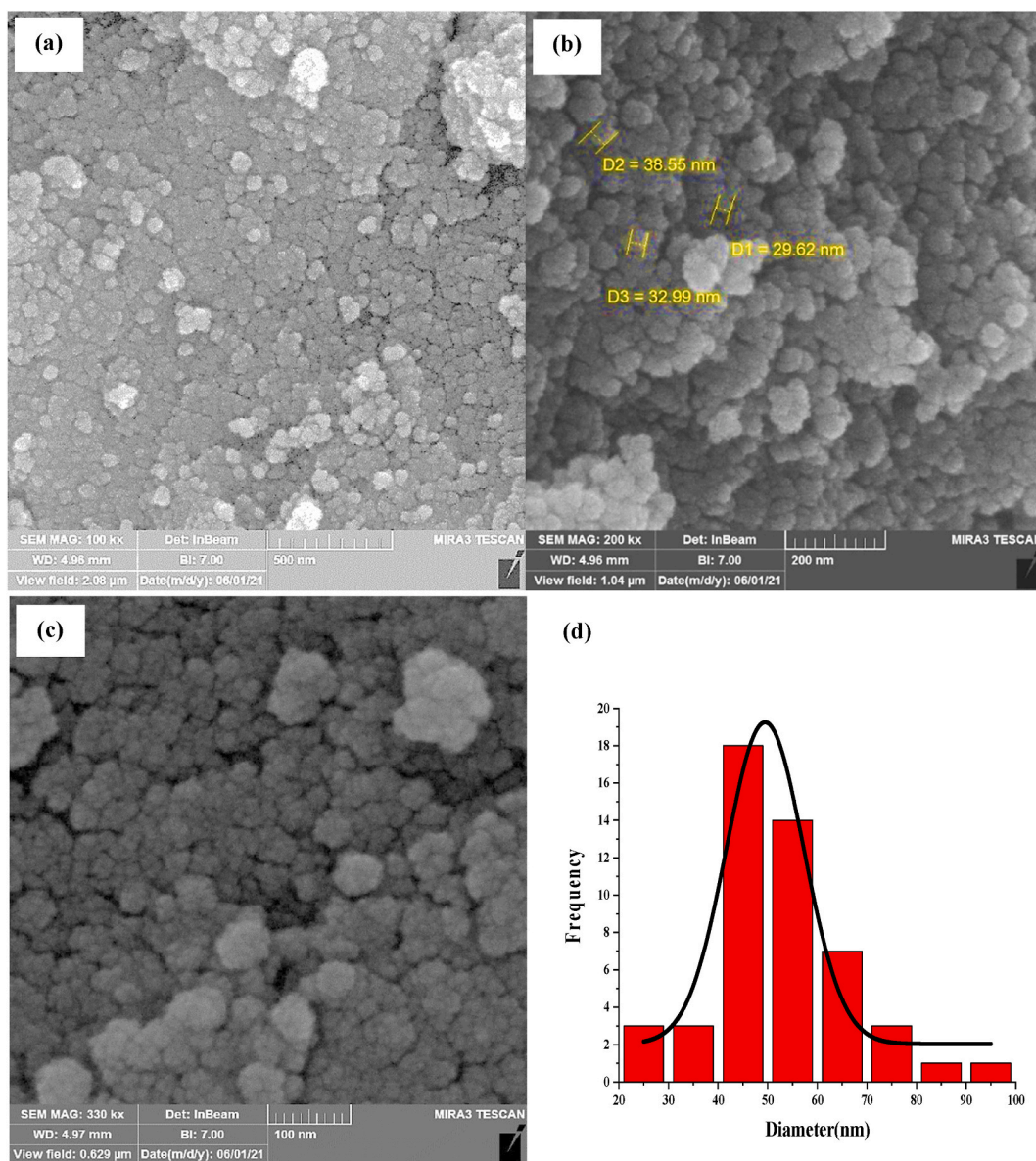


Fig. 2. FESEM images of ZnCoFe₂O₄@Ch magnetic nanoadsorbent with 500 nm (a), 200 nm (b), 100 nm (c) magnification and particle size distribution (d).

V: volume of solution (L)
m: adsorbent mass (g).

2.5. Determination of pH_{zpc}

To calculate the pH_{zpc} , first, 300 mL of KCl solution (0.1 M) was prepared with distilled water. Then the prepared solution was divided into six 50 mL containers. In each container, 0.01 g of nano-adsorbent was added after the pH of the solutions was adjusted to 2, 4, 6, 8, 10, and 12. The ultimate pH of these solutions was measured after they had been shaken on the shaker for 24 h. The intersection of ΔpH with the X axis, which was determined by drawing a graph of ΔpH (the difference between primary and secondary pH), and initial pH, was determined to be pH_{zpc} [7].

3. Results and discussion

3.1. Characterization of the synthesized magnetic nano adsorbent

Initially, the novel magnetic nano adsorbent $ZnCoFe_2O_4@Ch$ was prepared from Zn^{2+} , Co^{2+} , and Fe^{3+} in the presence of chitosan by using a microwave-assisted technique based on the experimental procedure. To characterize this magnetic nano adsorbent, several techniques were applied. To determine the shape and size, the FESEM images of the produced $ZnCoFe_2O_4@Ch$ were illustrated. Fig. 2 (a–c) depict the FESEM images of $ZnCoFe_2O_4@Ch$ synthesized in the presence of chitosan as a polysaccharide. As evident from the figure, the magnetic nanoadsorbent formed in the presence of chitosan exhibited loosely aggregated, evenly distributed, and gently rounded morphology. Fig. 2d shows the particle size distribution. From the particle size distribution histogram, it can be concluded that the average particle size of magnetic nano adsorbent is 40–50 nm. Rahmi et al. also synthesized nanocomposites in the presence of chitosan, which was highly efficient in removing cadmium. They discovered that the syringe-drop-wising procedure was mostly responsible for influencing the form of nanoparticles. Further evidence for an equitable distribution of Fe_3O_4 was provided by the presence of particle-like fillers that covered the whole surface of the bead [59–61].

The purity and chemical structure of synthesized $ZnCoFe_2O_4@Ch$ were evaluated by EDS analysis (Fig. 3). The EDS results demonstrate 44.24 % O, 27.33 % Fe, 8.13 % C, 7.75 % Co, 7.25 % Zn, and 5.31 % N in the chemical structure of $ZnCoFe_2O_4@Ch$ magnetic nano adsorbent which are following the expected values.

Mapping was used to investigate the $ZnCoFe_2O_4@Ch$ magnetic nano adsorbents element distribution (Fig. 4a). The green, blue, purple, turquoise, red, and yellow spots in Fig. 4 illustrate the distribution of elements on the surface of nanoparticles; these hues correspond to the elements oxygen, nitrogen, iron, cobalt, carbon, and zinc, respectively. Based on the obtained results, Co, Fe, O, N, Zn, and C had a homogeneous distribution that indicated high uniformity of the synthesized $ZnCoFe_2O_4@Ch$. Also, line scanning shown in Fig. 4b has confirmed these results.

Fourier transform infrared spectroscopy (FTIR) has been used to identify the functional group of synthesized nanoparticles. By using a KBr disc, the powdered $ZnCoFe_2O_4@Ch$ was investigated (Fig. 5). In the FTIR spectrum, chitosan absorption bands are shown (Fig. 5). The stretching vibrations of the hydroxyl group O–H are responsible for the absorption band in the 3418 cm^{-1} range. The stretching vibrations of the C–H group are associated with the absorption band in the range of 2866 cm^{-1} . The bending vibrations of the primary amine group NH_2 and the stretching vibrations of the C–O alcoholic group were observed in the range of 1621 cm^{-1} and 1160 cm^{-1} , respectively. In the ranges of 1121 cm^{-1} and 2001 cm^{-1} , respectively, etheral C–O stretching vibrations and C–N stretching vibrations were seen [62–64]. Additionally, according to the FTIR spectra, the absorption bands for $ZnCoFe_2O_4@Ch$

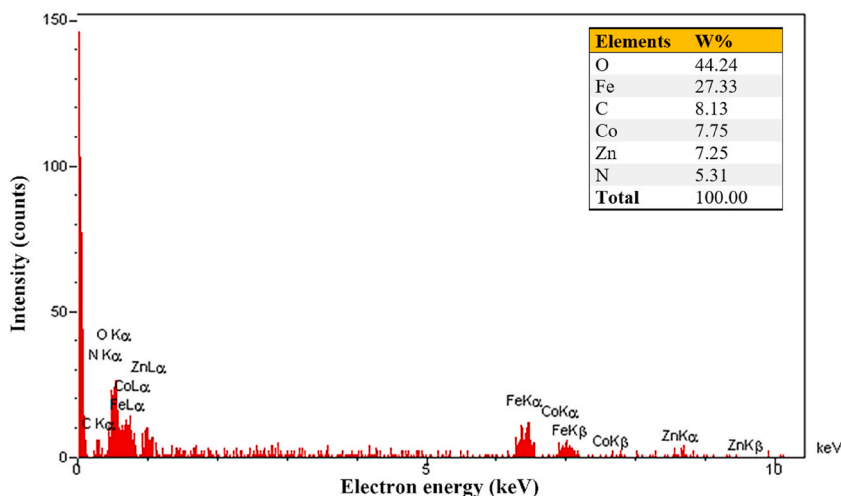


Fig. 3. EDS of $ZnCoFe_2O_4@Ch$ magnetic nanoadsorbent.

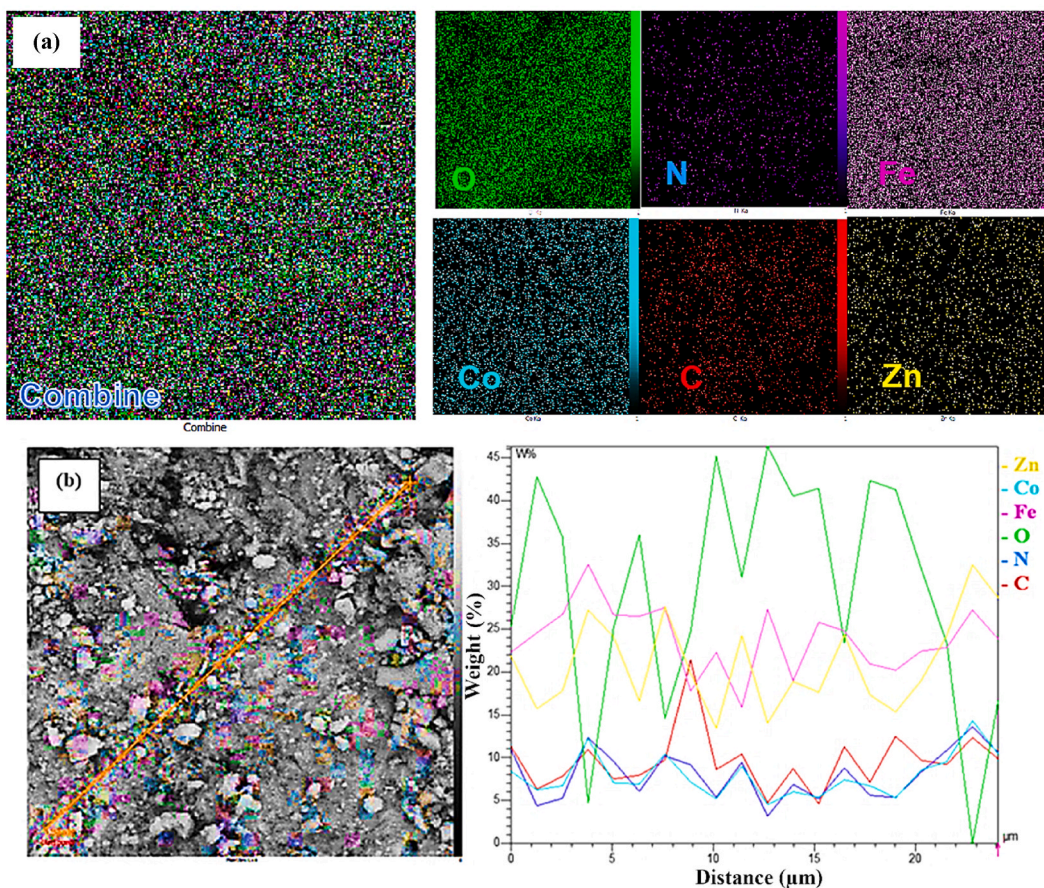


Fig. 4. Mapping (a) and line scanning (b) of ZnCoFe₂O₄@Ch magnetic nanoadsorbent.

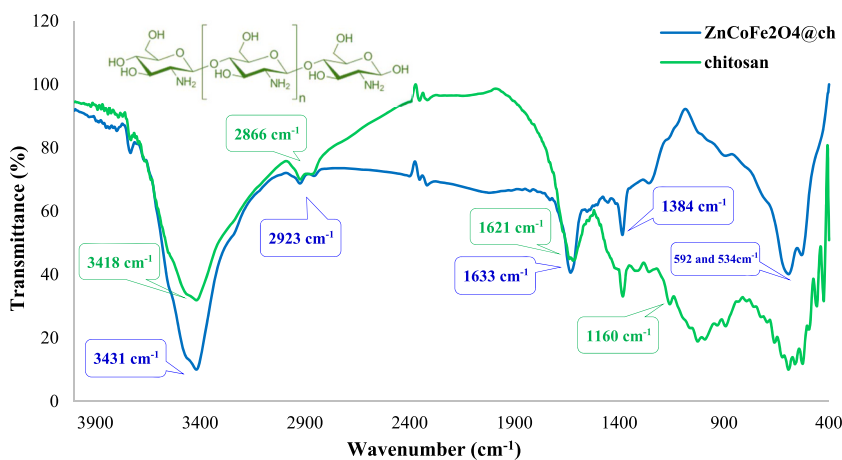


Fig. 5. FTIR spectra of ZnCoFe₂O₄@Ch magnetic nanoadsorbent and Chitosan.

nanosorbents were as follows:

Stretching vibrations of the O-H group were found in the range of 3431 cm⁻¹, stretching vibrations of the methylene -CH₂ group in the range of 2923 cm⁻¹, bending vibrations of the NH₂ amine group in the range of 1633 cm⁻¹, and C-N stretching vibrations in the range of 1384 cm⁻¹ [62,65]. Additionally, the ferrite spinel ZnCoFe₂O₄@Ch has two absorption bands in the range of 592 and 534 cm⁻¹ that are related to metal oxides in the octahedral and quadrilateral structures, respectively [66,67]. These findings support the effective synthesis of the nano sorbent ZnCoFe₂O₄@Ch in the presence of chitosan.

The XRD patterns of Chitosan, $\text{ZnCoFe}_2\text{O}_4$, and $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nano-adsorbent are shown in Fig. 6a-c. XRD analysis was used to characterize structures, phases, and crystalline composition of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$. The achieved results are shown in Fig. 6. The XRD pattern of Chitosan presents a prominent peak centered at 19.67° . Based on the JCPDS 96-591-0064, the $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ crystal phase structure and XRD pattern with diffraction peaks at $2\theta = 35.47^\circ$, 43.12° , 57.37° , 62.87° and 67.87° are indexed to the $\text{ZnCoFe}_2\text{O}_4$ cubic spinel phase. According to the obtained results, the $\text{ZnCoFe}_2\text{O}_4$ crystal structure was well-formed. It is so interesting that the XRD spectrum of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ indicates the characteristic peaks linked with chitosan and $\text{ZnCoFe}_2\text{O}_4$, which points to the successful synthesis of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nanoadsorbent [59–61].

Fig. 7 shows adsorption/desorption isotherm, BET-BJH specific surface area, and t-plot of the synthesized $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ (Fig. 7a–d). According to the BET plot the specific surface area, mean pore diameter, and total pore volume of synthesized magnetic nano adsorbent were obtained at $85.65 \text{ m}^2/\text{g}$, 6.86 nm , and $0.14 \text{ cm}^3/\text{g}$, respectively. $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ could be classified as a mesoporous material [68]. Also, the high specific surface area of the nano-adsorbent ($85.65 \text{ m}^2/\text{g}$) causes more interaction with the TC as a pollutant, which will increase the removal efficiency of the process. The literature also confirms these results [59–61].

By using a vibrating sample magnetometer, the magnetic properties of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nano adsorbent were examined (Fig. 8a). The remnant magnetization (M_r), saturation magnetization (M_s), and coercive force (H_c) values were obtained 6.29 emu/g , 48.03 emu/g , and 50 Oe , respectively. These values depict the high magnetic strength of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nano adsorbent. The results demonstrate that the $\text{ZnCoFe}_2\text{O}_4$ magnetic property in the $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ structure is completely preserved.

TGA analysis was carried out to evaluate the thermal stability of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ in the temperature range 25°C – 800°C , and results are depicted in Fig. 8b. The first mass loss step happened in the temperature range of 25°C up to about 200°C which is linked with the adsorbed water loss (-1.46%). Due to the breakdown of Ch glycosidic bonds and also, its decomposition into lower fatty acids, the weight loss was reported at 200 – 800°C (-3.82%). According to the results, the adsorbent total weight loss was seen as 5.28% until 800°C which shows the $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ high thermal stability. The TGA results of this research were confirmed by previous research that used nanocomposites of chitosan as adsorbents [59–61].

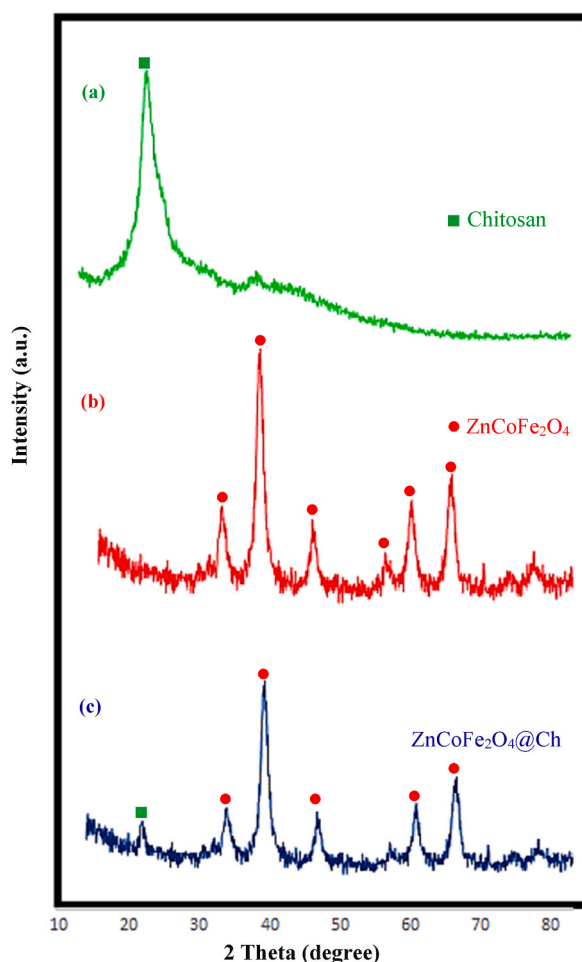


Fig. 6. The XRD patterns of Chitosan (a), $\text{ZnCoFe}_2\text{O}_4$ (b), and $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ (c) magnetic nanoadsorbent.

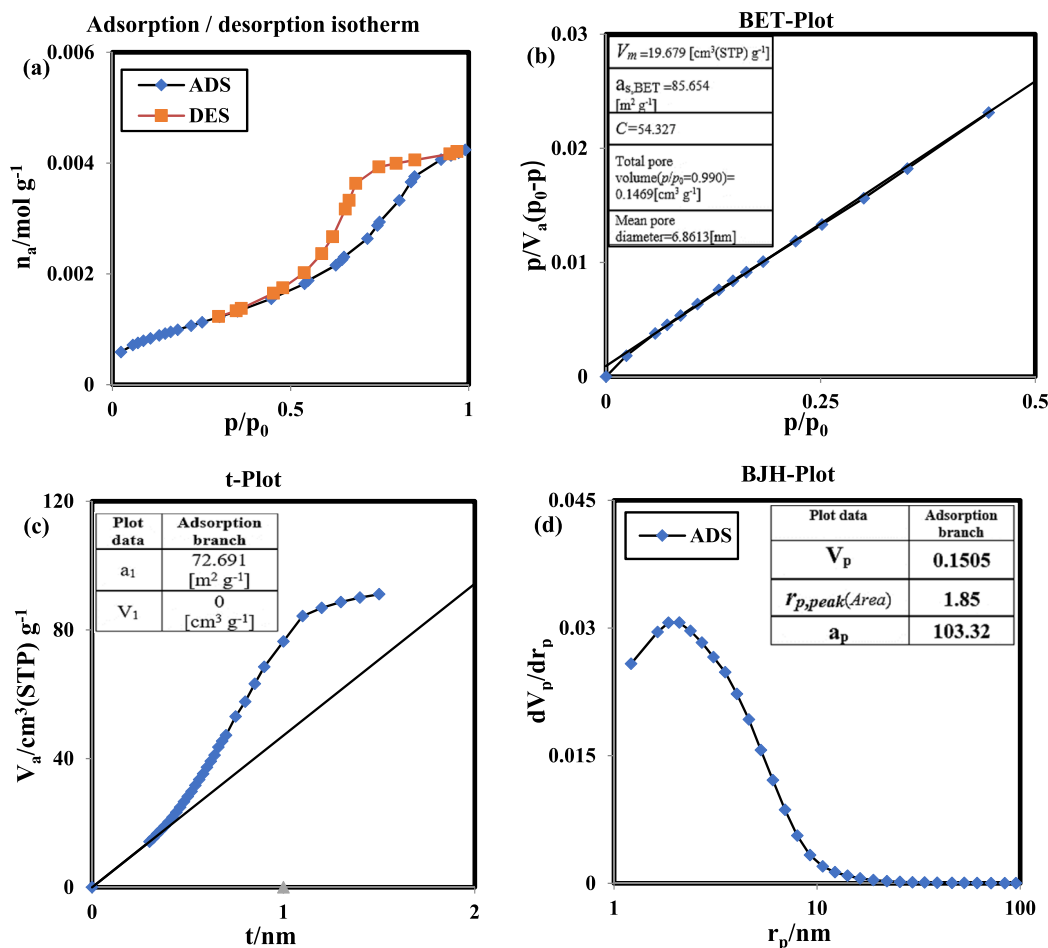


Fig. 7. Adsorption/desorption isotherm (a), BET surface area (b), t-Plot (c), and BJH surface area (d) of ZnCoFe₂O₄@Ch magnetic nanoadsorbent.

3.2. Adsorption test optimization

3.2.1. Effect of adsorbent dose

Fig. 9 (a) illustrates the impact of the nanosorbent's quantity on the effectiveness of TC removal after 45 min. The initial TC concentration in this method was 10 mg/L, the initial solution volume was 100 mL, and several doses of nanosorbents (0.7, 1, 1.5, and 2 g/L) were examined. The adsorption efficiency of TC was enhanced from 77 % to 85 % by increasing the quantity of nano sorbent from 0.7 to 1 g/L. The removal effectiveness improved by 93 % at higher nano sorbent doses of 2 g/L, but because this rise in removal efficiency was not evident and was not cost-effective, the optimal dose of 1 g/L nano sorbent with 85 % removal efficiency was taken into consideration. The increased contact area of the adsorbent and the availability of additional adsorption sites might have contributed to an increase in removal effectiveness with increasing adsorbent dosage. On the other hand, since the concentration of TC was constant and the amount of interaction between the antibiotic molecules and the adsorbent nanoparticles did not greatly rise, its adsorption rate did not dramatically increase with increasing adsorbent dosage. In the process of removing TC using magnetic nanosorbents (Fe-MCM-41-A), Guo et al. observed that by increasing the dose from 0.1 g/L to 1 g/L, the removal efficiency increased from 47 % to 98 %. This observation is consistent with the findings of this study and can be attributed to the increase in the level of adsorbent contact [69].

3.2.2. Effect of pH

In the adsorption process, the impact of pH is substantial. Under the premise that the initial concentration of TC was 10 mg/L and the ideal quantity of nano sorbent (1 g/L), the results of the influence of pH on the removal effectiveness of TC by magnetic nanocomposite are demonstrated in a period of 2–45 min (Fig. 9b). As the pH was 3, the maximum removal effectiveness was 85 %, and when the pH rose to 11, it fell to 9 %. As a result, 3 was considered to be the optimum pH. On the other hand, the adsorbent pH_{zpc} also has an impact on the effectiveness of the adsorption process. Similar to this, the pH_{zpc} of ZnCoFe₂O₄@Ch nano sorbent was measured, and its value was reported to be 6.8 (Fig. 9d). According to the pH_{zpc} value for the nano sorbent, it was revealed that the adsorbent

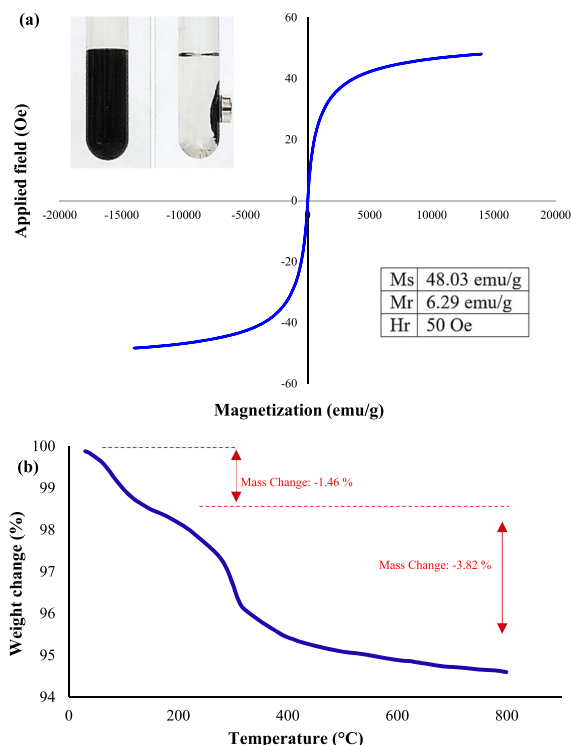


Fig. 8. VSM magnetization curve of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nano adsorbent; the photo inset shows the solution after magnetic separation of the magnetic nano adsorbent and (a) the TGA pattern of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ (b).

surface is positive at pHs below 6.8 and negative at pHs over 6.8. Given that TC has three acid dissociation constants (pKa) of 3.3, 7.68, and 9.68, as shown in Fig. 9c, it can exist in a variety of ionic forms at various pH levels. Between pH 3.3 to 7.68, TC molecules may be found in zwitterion or neutral form, at pH 3.3 can be found in cationic form, and above pH 7.68, they can be found in the anionic form [70]. Since the optimum pH for this method is claimed to be 3, the presence of the tetracycline molecule in cationic form and the positive charge on the adsorbent surface means that electrostatic attraction forces won't be a major factor in the adsorption of pollutants on the adsorbent surface at this pH. Therefore, TC tends to separate from the aqueous medium under the influence of lyophobic forces and reach the surface of the adsorbent under these circumstances because it is an organic compound in the aqueous medium. As a result, this mechanism in the adsorption of TC on the adsorbent was predominant [71]. Luo et al.'s study on the adsorption of TC through modified alginate grains revealed that a pH of 5 was ideal for the process. Thus, compared to neutral and alkaline pHs, acidic pHs had the best effectiveness, which is consistent with the findings of this study [72].

3.2.3. Effect of the initial concentration

Tetracycline solutions at concentrations of 5, 10, 15, 20, and 30 mg/L, with an optimum pH of 3, and a temperature of 25 °C were in contact with 1 g/L of $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$ magnetic nano sorbent to examine the impact of variations in TC concentration on its removal effectiveness (Fig. 9e). According to the findings of concentration changes, the adsorption effectiveness of TC declined from 90 % to 67 % when the initial concentration of TC solution was increased from 5 mg/L to 30 mg/L. As a consequence, the ideal concentration of TC in the process was 5 mg/L. Since the solution's TC molecule density rises with the initial concentration, the active sites on the adsorbent surface are occupied more quickly, and the effectiveness of TC removal declines [73]. According to Kasraei et al.'s work on the removal of TC from wastewater using the adsorption process ($\text{Fe}_3\text{O}_4@\text{TOMATS IL}$), the removal efficiency fell as the concentration of TC rose, which is consistent with the findings of this investigation [74].

3.2.4. Effect of temperature

Another factor influencing the adsorption process is the temperature. Fig. 9 (f) depicts the influence of temperature fluctuations on the adsorption efficiency of tetracycline. At a concentration of 5 mg/L of TC and pH 3, the findings of the investigation on the influence of process temperature (25, 35, 45, and 55 °C) on the removal efficiency of TC by magnetic nano sorbent revealed that at 25 °C temperature, 90 % of the antibiotic was effectively adsorbable. Tetracycline's removal effectiveness dropped to 86 % at 55 °C due to a reduction in removal rate as the temperature rose to that point. This outcome demonstrates the exothermic nature of the adsorption process as well as physical adsorption. Because raising the temperature did not improve efficiency, chemical adsorption did not take place; instead, physical adsorption occurred. van der Waals forces had a stronger impact on physical adsorption, which increased the adsorption efficiency [7]. The trend of negative ΔH values in Ahmad et al.'s research for the removal of TC by poly

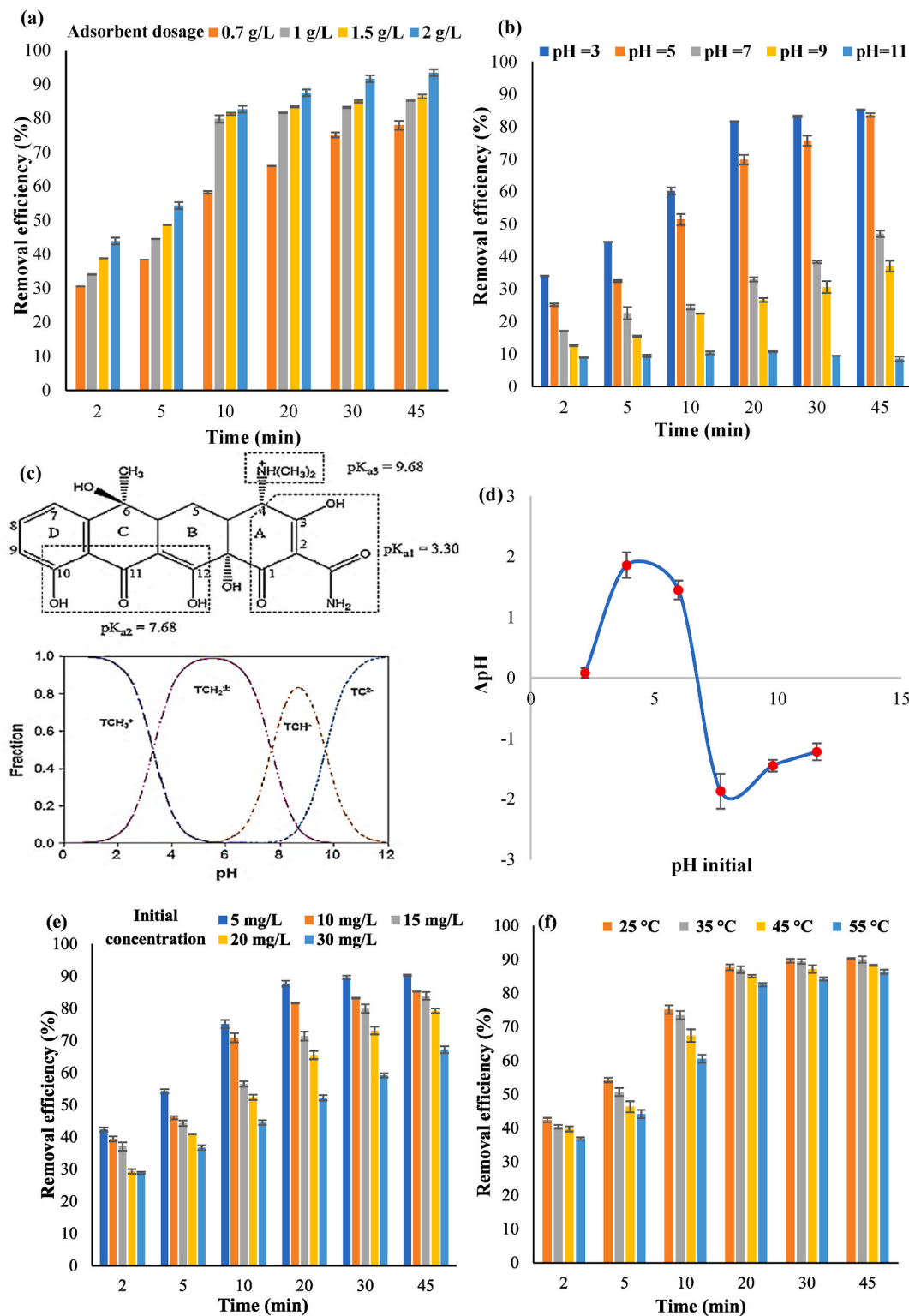


Fig. 9. Effect of adsorbent dose (a), pH (b), the ionic forms of TC at different pHs (c), pH_{zpc} (d), TC concentration (e), and temperature (f) (optimum condition: pH: 3, nano-adsorbent dosage: 1 g/L, TC initial concentration: 5 mg/L, and temperature: 25 °C).

(cyclotriphosphazene-co-4,4'-sulfonyldiphenol)/rGO suggested that the adsorption process was exothermic, which is compatible with the results of the current study [75].

3.3. Adsorption isotherm models

In this process, the equilibrium state of the adsorbate on the adsorbent surface was studied using the Langmuir and Freundlich isotherms. According to the Langmuir model (Eq. (3)), adsorption occurs as a single layer, with the adsorbate molecule on the adsorbent surface and constant adsorption energy. The adsorption of one molecule at a site does not impact the adsorption of other molecules at the same site [76].

$$\frac{C_e}{Q_e} = \frac{1}{Q_{\max} K_L} + \frac{C_e}{Q_{\max}} \quad (\text{Eq. 3})$$

The maximum adsorption capacity is represented in this equation by $Q_{\max}$ (mg/g). The equilibrium concentration of TC is expressed as C_e (mg/L), the quantity of the adsorbed substance is expressed as Q_e (mg/g), and Langmuir's adsorption equilibrium constant is expressed as K_L (L/mg).

R_L (Eq. (4)), which studies the state of the adsorption process, is another parameter. Thus, the adsorption process is irreversible if $R_L = 0$, the favorable adsorption process if $1 > R_L > 0$, the linear adsorption process if $R_L = 1$, and the unfavorable adsorption process if $R_L > 1$ [77].

$$R_L = \frac{1}{1 + K_L C_0} \quad (\text{Eq. 4})$$

where tetracycline's initial concentration in mg/L is represented by C_0 in this equation.

The Freundlich model (Eq. (5)) may be utilized for heterogeneous surface adsorption as well as multilayer adsorption with heterogeneous heat distribution.

$$\log q_e = \log K_F + \frac{1}{n} \log C_e \quad (\text{Eq. 5})$$

Freundlich's adsorption equilibrium constant is K_F ((mg/g) (L/mg)^{1/n}) in this equation. The kind of isotherm is indicated by the heterogeneity factor, also known as surface adsorption intensity or $1/n$. In other words, the adsorption process is reversible if $1/n = 0$, favorable if $0 > 1/n > 1$, and unfavorable if $1 > 1/n$ [78]. Fig. 10 (a, b) shows the findings from studied Langmuir and Freundlich equilibrium isotherms.

The contacts between the adsorbent and the adsorbate are taken into consideration by the Temkin isotherm. Under Temkin isotherm, the temperature of adsorption in several layers declines logarithmically as a result of the reciprocal impacts of the adsorbent and the adsorbate on one another. Equations (6) and (7) display the Temkin isotherm's linear formula.

$$Q_e = BT \ln K_t + BT \ln C_e \quad (\text{Eq. 6})$$

$$bT = (RT) / (BT) \quad (\text{Eq. 7})$$

As BT and bT are constants in these equations, K_t stands for the Temkin adsorption potential (L/g). T is the absolute temperature in Kelvin, and R is the universal gas constant (8.314 J/mol.K).

To consider the impact of the porous structure of adsorbents, the Dubinin-Radushkevich isotherm model is utilized. It clarifies the porosity and adsorption energy of the adsorbent. The linear formula for the Dubinin-Radushkevich isotherm is shown in equations (8) and (9) [1,3].

$$\ln q_e = \ln q_m - \beta \cdot e^2 \quad (\text{Eq. 8})$$

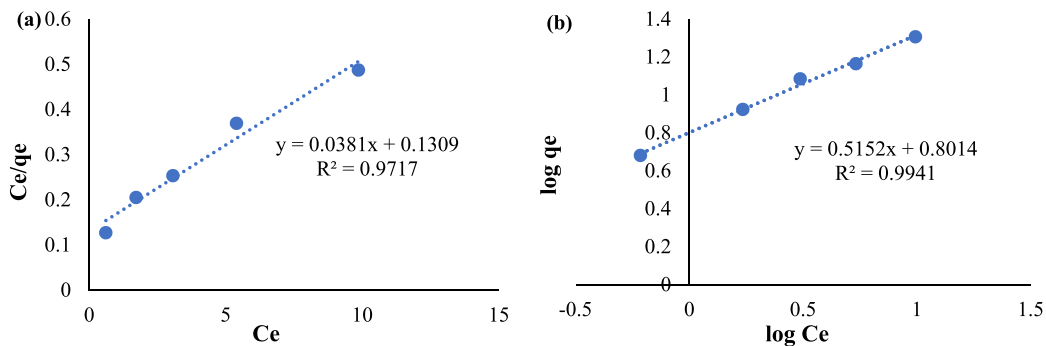


Fig. 10. Langmuir (a) and Freundlich (b) isotherm plot (adsorbent: 1 g/L, contact time: 30 min, pH: 3, and temperature: 25 °C).

$$\varepsilon = RT \ln \left(\frac{C_0}{C_e} \right) \quad (\text{Eq. 9})$$

The adsorption energy is represented by the constant β (mol^2/kJ^2), while the adsorption potential is represented by ε (kJ/mol).

Table 1 displays the results from studied Freundlich, Langmuir, Temkin, and Dubinin-Radushkevich equilibrium isotherms.

The findings shown in Table 1 imply that the Freundlich model's R^2 value ($R^2 = 0.994$) was higher than the Langmuir model's R^2 value ($R^2 = 0.972$). The Freundlich isotherm model is, therefore, more fitted to describe the adsorption process. Additionally, the Freundlich model's value of $1/n$ lies between 0 and 1, suggesting that adsorption is preferable. The results of this work are compatible with Luo et al.'s isotherm study for the adsorption of TC from aqueous solutions by modified alginate beads, which demonstrated that the adsorption process of TC follows the Freundlich isotherm [72].

3.4. Kinetic study

To comprehend the dynamics and the impact of variables impacting the rate of tetracycline adsorption on the adsorbent, adsorption kinetics was studied. In this investigation, model kinetics of *pseudo*-first-order (Eq. (10)), *pseudo*-second-order (Eq. (11)), and intra-particle kinetic model (Eq. (12)) were employed.

$$\ln(q_e - q_t) = \ln q_e - K_1 t \quad (\text{Eq. 10})$$

The *pseudo*-first-order model's speed constant is K_1 (min^{-1}) in this equation. Tetracycline adsorbed amounts are shown in q_e (mg/g) and q_t (mg/g), respectively, for the equilibrium state and given periods (min).

$$\frac{t}{q_t} = \frac{1}{K_2 q_e^2} + \frac{1}{q_e} \times t \quad (\text{Eq. 11})$$

where K_2 ($\text{g/mg} \cdot \text{min}$) is the *pseudo*-second-order kinetic constant in this equation. Tetracycline adsorbed amounts are shown in q_e and q_t (mg/g), respectively, at equilibrium and at any given time (min). The slope of the linear graph of the equation may be used to determine the speed constant K_2 [79]. Fig. 11 (a, b) shows the findings from the studied *pseudo*-first-order and *pseudo*-second-order kinetic models.

The restricting phase that regulates the kinetics in the majority of adsorption processes is frequently thought of as intraparticle diffusion kinetics. The quantity of absorbance against the square of time, as represented by Weber and Morris, is plotted to determine the likelihood of kinetic limiting by transport in pores [80].

$$q_t = K_i t^{0.5} + C \quad (\text{Eq. 12})$$

where K_i ($\text{g/mg} \cdot \text{min}$) in this correlation stands for the particle's internal diffusion rate constant and C for the boundary layer's penetration or surface adsorption.

The Elovich equation is appropriate for systems with heterogeneous adsorbing surfaces and can be met by chemical adsorption processes, which describe many pollutant adsorption systems. The following equation (Eq. 13) serves as a foundation for calculating this kinetic model.

$$q_t = 1/\beta \ln(\alpha \cdot \beta) + 1/\beta \ln t \quad (\text{Eq. 13})$$

The initial adsorption rate constant in this equation is α ($\text{mg/g} \cdot \text{min}$), while the desorption rate constant is β (g/mg) [81].

Table 2 reports the findings of the kinetic studies of the TC adsorption process by $\text{ZnCoFe}_2\text{O}_4/\text{Ch}$.

The *pseudo*-second-order model is used to describe the adsorption behavior since the value of R^2 in the *pseudo*-second-order kinetic model ($R^2 = 0.996$) is greater than the value of R^2 in the *pseudo*-first-order model ($R^2 = 0.467$). Therefore, TC adsorbed effectively on magnetic nano-adsorbent $\text{ZnCoFe}_2\text{O}_4/\text{Ch}$ using a *pseudo*-second-order kinetic model. In agreement with the findings of this investigation, Zhang et al.'s kinetic assessment for the adsorption of TC using $\text{CoFe}_2\text{O}_4/\text{MMT}$ magnetic nano-adsorbent revealed that the kinetics of the adsorption process followed *pseudo*-second-order kinetics [82].

Table 1

Langmuir, Freundlich, Temkin, and Dubinin-Radushkevich isotherm parameters (adsorbent: 1 g/L, contact time: 30 min, pH: 3, and temperature: 25 °C).

Langmuir				Freundlich			Temkin			Dubinin-Radushkevich		
R^2	R_L	$Q_{\max}$ (mg/g)	K_L	R^2	K_F	$1/n$	R^2	B_1	K_T (L/mg)	$Q_{\max}$ (mg/g)	E (kJ/mol)	R^2
0.972	0.408	26.241	0.291	0.994	6.321	0.515	0.963	5.419	3.289	7.295	31.623	0.827
	0.256											
	0.187											
	0.147											
	0.103											

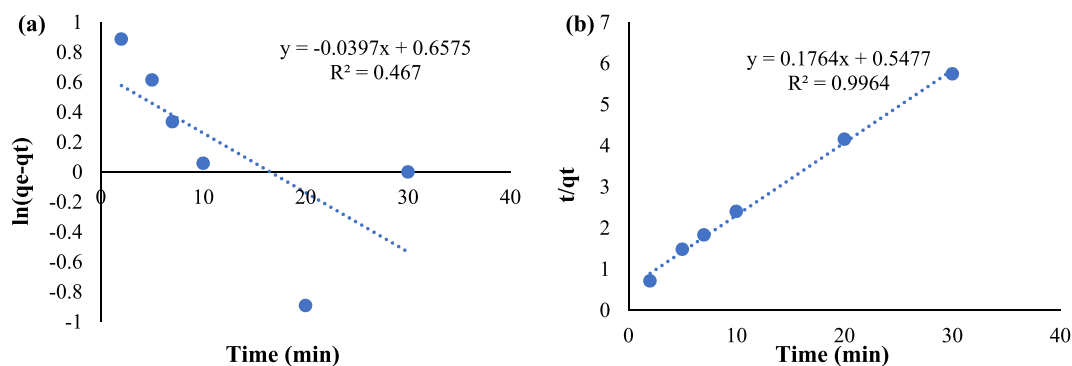


Fig. 11. Pseudo-first-order (a) and pseudo-second-order (b) kinetic plot (TC: 5 mg/L, Adsorbent: 1 g/L, pH: 3, and temperature: 25 °C).

Table 2

Coefficients for the adsorption kinetics models (TC: 5 mg/L, adsorbent: 1 g/L, pH: 3, and temperature: 25 °C).

Pseudo-first-order kinetic model			Pseudo-second-order kinetic model			Intraparticle kinetic model			Elovich kinetic model		
R ²	K ₁ (1/min)	q _e (mg/g)	R ²	K ₂ (g/mg.min)	q _e (mg/g)	R ²	K _p (mg/g. min ⁻⁵)	C (mg/g)	R ²	β (g/mg)	α (mg/g.min)
0.467	0.040	1.930	0.996	0.057	5.661	0.975	0.594	2.108	0.991	1.088	8.516

3.5. Thermodynamic study

Thermodynamic characteristics, such as variations in Gibbs free energy (ΔG), enthalpy (ΔH), and entropy (ΔS), may be studied using variations in the equilibrium constant with temperature in the equilibrium state. Equation (14) was used to determine how the adsorption process's Gibbs free energy fluctuates.

$$\Delta G = -RT \ln k_d \quad (\text{Eq. 14})$$

in this equation, ΔG is the Gibbs free energy change (KJ/mol), R is the general constant of gases (J/mol. K), and T is the temperature (K). Equation (15) was also used to find the equilibrium constant (K_d).

$$K_d = \frac{(C_0 - C_e)}{C_e} \times \frac{V}{W} \quad (\text{Eq. 15})$$

in this equation, the initial and equilibrium concentrations are C_0 (mg/L) and C_e (mg/L), respectively, as well the volume of the solution is V (L) and the mass of the adsorbent is W (g). The values of the standard enthalpy changes (ΔH) and standard entropy changes (ΔS) can be obtained by drawing the *Van't Hoff* plot, which is $\ln K_d$ versus $1/T$ (Fig. 12), and then determining the equation of the line and gain enthalpy and entropy changes values (Eq. (16)) [77].

$$\ln K_d = \frac{\Delta S}{R} - \frac{\Delta H}{RT} \quad (\text{Eq. 16})$$

Table 3 summarizes the findings of thermodynamic studies of the TC adsorption process on the $\text{ZnCoFe}_2\text{O}_4@\text{Ch}$.

The findings are compatible with Yu et al.'s study on the adsorption of TC antibiotic through magnetic graphene oxide, where the negative Gibbs free energy of the process shows that the adsorption process is carried out spontaneously [83]. The exothermic character of the adsorption process is shown by the negative of the typical enthalpy changes. Additionally, the negative standard

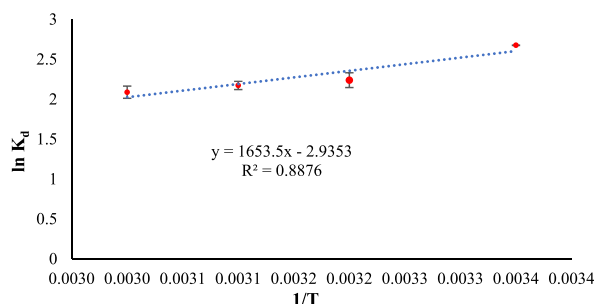


Fig. 12. Van't Hoff plot (pH: 3, nano adsorbent dosage: 1 g/L, TC initial concentration: 5 mg/L).

Table 3

Thermodynamic parameters of TC adsorption onto ZnCoFe₂O₄@Ch (pH: 3, nanoadsorbent dosage: 1 g/L, TC initial concentration: 5 mg/L).

T (k)	ΔG (kJ/mol)	ΔH (kJ/mol)	ΔS (j/mol.k)
298	−6.621	−15.162	−28.694
308	−5.730		
318	−5.742		
328	−5.693		

entropy changes show less disorder in the adsorption process. These findings are in accordance with the research by Turku et al. on the adsorption of TC by silica, which used typical enthalpy and entropy changes [84].

3.6. Investigation of process efficiency on real wastewater

Tetracycline adsorption was also evaluated on a real wastewater sample from Kerman University of Medical Sciences under ideal circumstances (pH: 3, nano adsorbent dosage: 1 g/L, contact time: 30 min, temperature: 25 °C). Table 4 reports the findings of the physio-chemical quality of this effluent.

The real sample's maximum adsorption efficiency was attained at a rate of 80 %. The presence of interfering elements in the wastewater sample, such as anions and cations, prevents the adsorption of the TC pollutant on the adsorbent surface and competes with TC for adsorption on the adsorbent surface, resulting in a drop in efficiency in the real sample. The results of this study were similar to Nasiri et al.'s study [7].

3.7. Comparison of the ZnCoFe₂O₄@Ch performance with other adsorbents

In Table 5, the effectiveness of magnetic nanocomposite ZnCoFe₂O₄@Ch in removing medicinal substances was contrasted with that of other synthetic magnetic adsorbents. ZnCoFe₂O₄@Ch magnetic nanocomposite had greater removal effectiveness in less time and adsorbent dosage, as well as higher adsorption capacity, according to Table 5.

3.8. Chemical stability and reusability of ZnCoFe₂O₄@Ch

Recovery and reusing of the adsorbent are crucial to the adsorption process from an economic and environmental perspective. Due to its strong magnetic properties, the magnetic nanocomposite ZnCoFe₂O₄@Ch is easily separated from the process environment by a magnet and does not cause secondary environmental contamination. After separating the adsorbent from the process medium using a magnet, it was put in a furnace at 200 °C for 1 h to remove the pollutants adsorbed on the adsorbent in order to assess the reusability of the employed nano sorbent. Then, under ideal circumstances, five recovery cycles were used to evaluate its effectiveness [69]. The removal effectiveness of the magnetic nanocomposite ZnCoFe₂O₄@Ch fell from 93 % to 89 % during two cycles. As shown in Fig. 13a, the removal effectiveness was 65 % after five recovery cycles (TC: 5 mg/L, Adsorbent: 1 g/L). After five stages of usage and regeneration, the ANOVA analysis revealed that there is no significant difference ($p = 0.201$) between the various cycles of adsorbent regeneration, indicating the satisfactory functioning of the adsorbent. The pollutant's presence in adsorbent sites may be the cause of this drop in efficiency, so, the explanation for the decline in TC removal effectiveness after five cycles of recovery and reuse may be attributed to the irreversible occupancy of the active sites of the adsorbent. According to Weng et al.'s study, after three cycles of using magnetic nanosorbents to remove the antibiotics ofloxacin and pefloxacin, antibiotic adsorption was 50 % and 67 % efficient, respectively (76). By using XRD, FESEM, and VSM analysis, the chemical stability of the adsorbent after five cycles of reuse and recovery was evaluated (Fig. 13b, c, and d). According to the XRD, FESEM, and VSM analysis, the nano adsorbent's crystal structure, morphology, and magnetic strength are still retained after five steps of recovery and may be utilized again to treat wastewater that contains TC. Also, the morphological properties of the adsorbent were still present after the fifth cycle, per the FESEM findings. There

Table 4

Physicochemical characteristics of real wastewater samples.

Parameter	Value
pH	7.35
SO ₄ (mg/L)	134.5
BOD ₅ (mg/L)	8
COD (mg/L)	26.1
TSS (mg/L)	78
TDS (mg/L)	1194
N (NH ₃) (mg/L)	1.4
TKN (mg/L)	1.82
PO ₄ orto (mg/L)	38.38
Tetracycline (mg/L)	5

Table 5Comparison of ZnCoFe₂O₄@Ch with other adsorbents.

No.	Adsorbent	Pollutant	Dose of adsorbent (g/L)	Concentration (mg/L)	Contact time (min)	Efficiency (%)		Recovery (%) \<	Adsorption capacity	Ref.
						Synthetic sample	Real sample \<			
1	Biochar (Rice straw)	Tetracycline	3	32.5	1440	92	–	–	13	[85]
2	Magnetic GAC	Metronidazole	1	20	90	92	–	–	–	[86]
3	AC-Fe ₃ O ₄	Amoxicillin	1	50	90	90	–	–	–	[87]
4	Biochar rice	Tetracycline	1.5	10	60	70	–	–	13.33	[88]
5	Biochar rice	Metronidazole	1.5	10	45	90	–	–	21.33	[88]
6	Magnetic iron nanoparticles (nFe)	Ofloxacin	0.2	5	300	100	56	67	12.8	[89]
7	ZnCoFe ₂ O ₄ @Ch	Tetracycline	1	5	30	93	80	65	26.24	This work

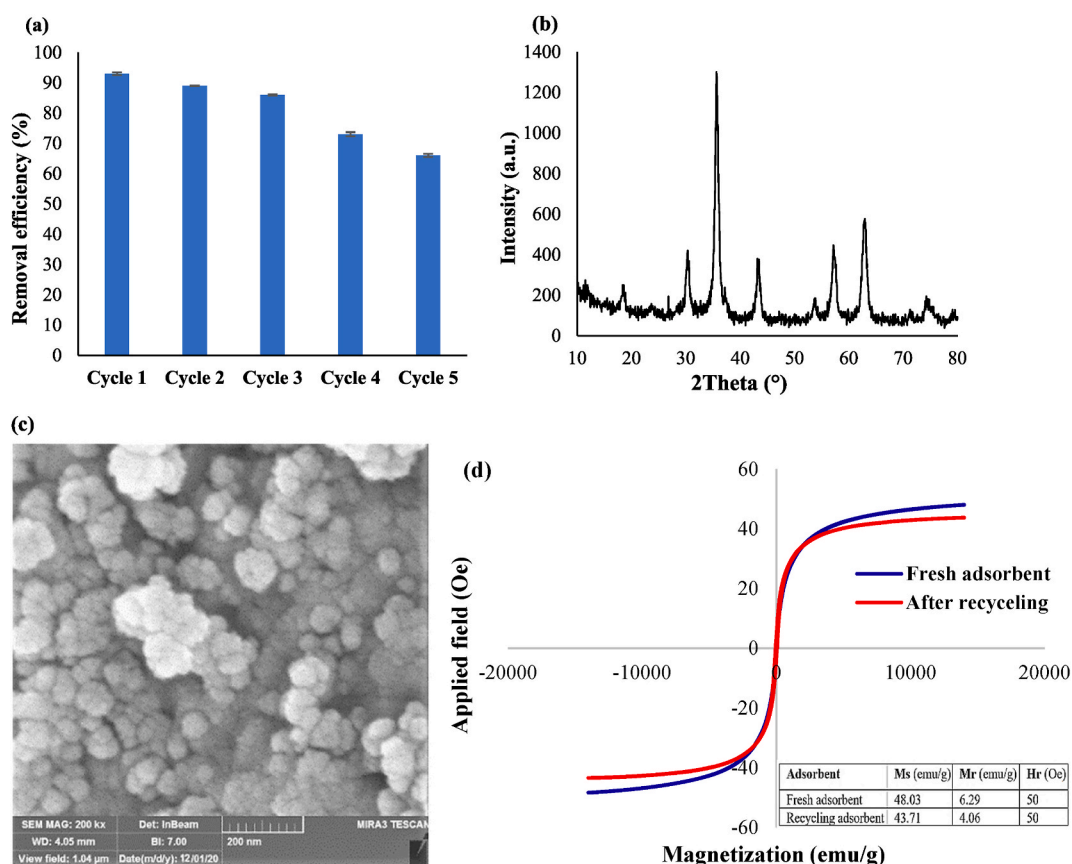


Fig. 13. Recycling of ZnCoFe₂O₄@Ch in the removal of TC (pH: 3, nano adsorbent dosage: 1 g/L, TC initial concentration: 5 mg/L) (a), XRD (b), FESEM (c) and VSM (d) analysis after five recovery steps.

was no discernible change in the location of 2 Theta in the XRD analysis pattern. Additionally, the XRD analysis's strong peaks show the adsorbent's great chemical stability.

3.9. Adsorption mechanism

The FTIR spectrum of the adsorbent before and after the adsorption process has been investigated to better understand how TC is adsorbed into the adsorbent surface (Fig. 14a). Following the adsorption process, the strength of several peaks has changed in the absorber's FTIR spectrum, shifting to a lower wavelength. This suggests that these groups took part in the TC adsorption process. It is clear from the spectra that the band around 3431 cm⁻¹ corresponding to the stretching vibrations of the hydroxyl functional groups is shifted to a lower wavelength of 3410 cm⁻¹, the band around 2866 cm⁻¹ corresponding to the stretching vibrations of the C-H group to a lower wavelength of 2848 cm⁻¹, the band around 1621 cm⁻¹ corresponding to the bending vibrations of amine group to a lower wavelength of 1614 cm⁻¹, and two absorption bands in the range of 592 cm⁻¹ and 534 cm⁻¹, associated to metal oxides in the octahedral and tetrahedral structure of ferrites spinel of nano adsorbent to lower wavelengths of 584 cm⁻¹ and 467 cm⁻¹, respectively, and their intensities were reduced after TC adsorption, suggesting the participation of these groups in the TC adsorption process.

The mechanisms that may influence TC adsorption by ZnCoFe₂O₄@Ch nanosorbents are discussed further below. Through various forces acting on the TC and the adsorbent during the TC adsorption process, these interactions may take effect. Different regions of the ZnCoFe₂O₄@Ch nano sorbent can interact with the π electrons of the aromatic ring of the TC molecule. The π electrons of the TC aromatic ring can establish a CH- π bond with the CH groups in the chitosan structure [90,91]. Therefore, the van der Waals attraction force between the π electrons of the aromatic ring of the TC molecule and the hydrogen atom of the CH group in chitosan is necessary for the adsorption of TC molecules by the matrix structure of chitosan to generate a CH- π bond. The formation of an NH₂/OH- π bond with the amine group or the hydroxyl of chitosan is another hypothesis that may be involved in the TC adsorption process [91]. Chitosan is a polar compound that tends to produce hydrogen bonds with hydrophobic substances like TC since it has amine and hydroxyl groups in its structure [91,92]. The OH and NH₂ groups in the chitosan structure can form hydrogen bonds with the TC molecule's organic and hydrophobic nature. Additionally, in the adsorbent structure of ZnCoFe₂O₄@Ch, metal ions of transition metals including cobalt, zinc, and iron can function as either Lewis's acids or electron receptors. Functional groups in the TC structure like hydroxyl, amine, and carbonyl may also act as electron donor groups (EDGs) for transition metals such as cobalt, zinc, and iron.

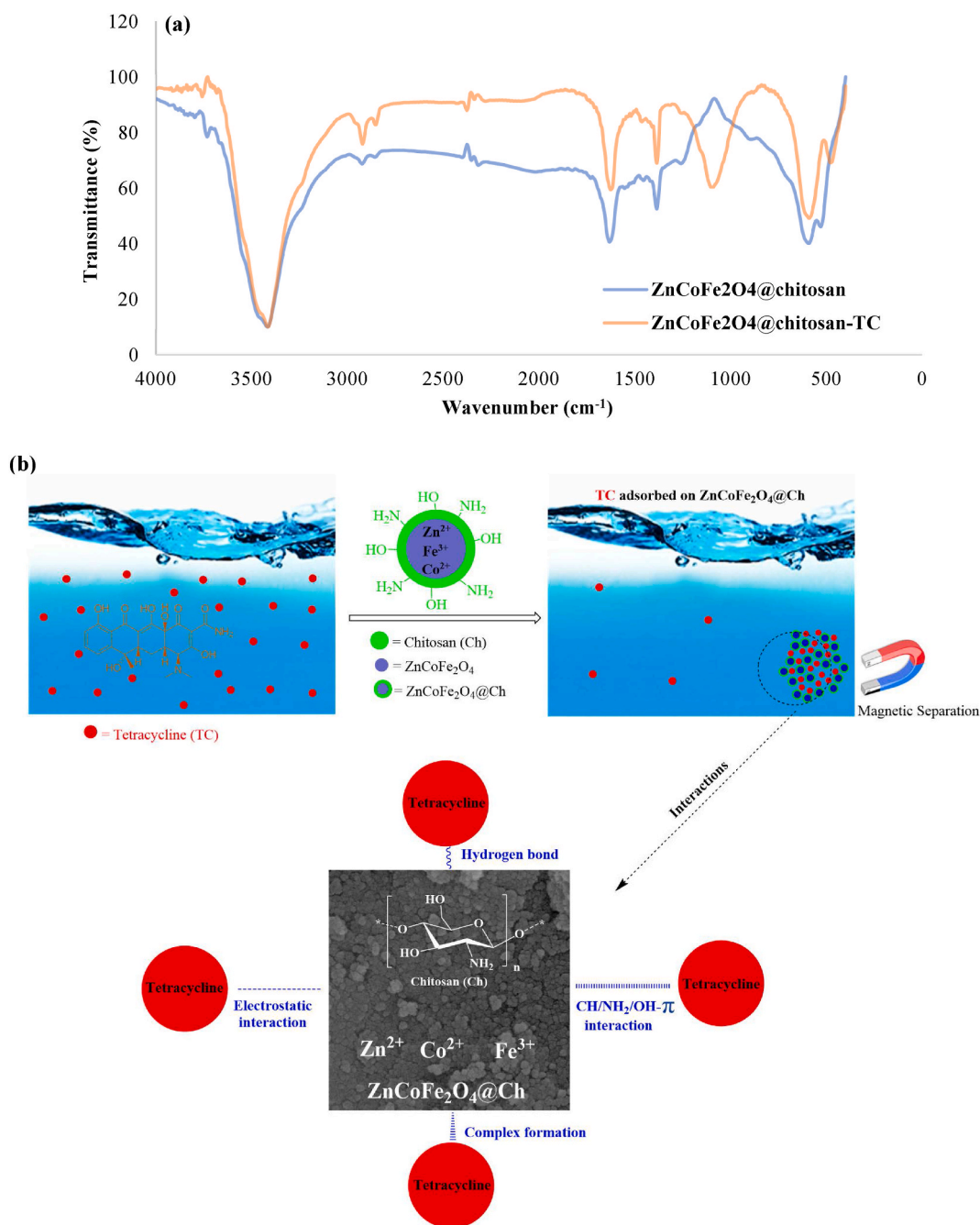


Fig. 14. FTIR of $\text{ZnCoFe}_2\text{O}_4@Ch$ before and after Tetracycline adsorption (a) and proposed mechanisms for TC adsorption on the $\text{ZnCoFe}_2\text{O}_4@Ch$ (b).

The EDGs of TC interact with the metal ions of the transition metals as electron-withdrawing groups (EWGs) to form a complex between the TC and the adsorbent. The development of this complex may have an impact on the process of TC adsorption [54,55]. Fig. 14b shows a schematic of the potential pathways for the $\text{ZnCoFe}_2\text{O}_4@Ch$ nanocomposite's TC adsorption process.

3.10. The performance comparison of $\text{ZnCoFe}_2\text{O}_4@Ch$, $\text{ZnCoFe}_2\text{O}_4$, and Ch in TC removal

Fig. 15 compares the TC adsorption efficiencies of $\text{ZnCoFe}_2\text{O}_4@Ch$, $\text{ZnCoFe}_2\text{O}_4$, and Ch under optimum adsorption conditions. The results showed that $\text{ZnCoFe}_2\text{O}_4@Ch$, $\text{ZnCoFe}_2\text{O}_4$, and Ch were able to remove 93 %, 82 %, and 65 % of TC, respectively. Compared to $\text{ZnCoFe}_2\text{O}_4@Ch$ and $\text{ZnCoFe}_2\text{O}_4$, chitosan was less effective in TC removal despite having functional groups. Also, the separation of Ch

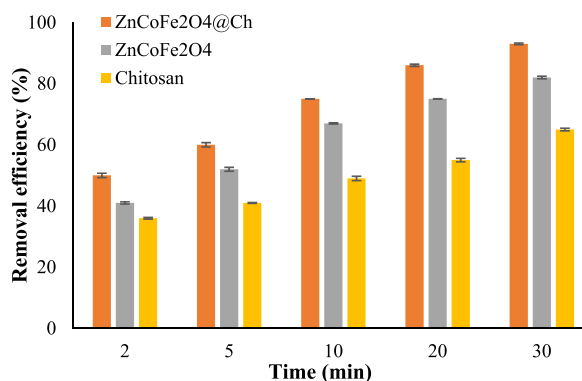


Fig. 15. The performance comparison of ZnCoFe₂O₄@Ch, ZnCoFe₂O₄, and Chitosan for TC adsorption (temperature = 25 °C, pH = 3, dosage nano-adsorbent = 1 g/L, concentration TC = 5 mg/L).

from the process solution should be done by filtration, which is very time-consuming. In addition, Ch is not in the nanoscale and has a smaller surface area compared to ZnCoFe₂O₄@Ch and ZnCoFe₂O₄, and its adsorption capacity is lower. ZnCoFe₂O₄ nanoparticles compared to Ch were more effective in removing TC. However, their efficiency was lower than ZnCoFe₂O₄@Ch. ZnCoFe₂O₄ nanoparticles have a larger surface area compared to Ch because they are on a nanoscale, and due to their magnetic properties, they are easily and quickly separated from the process solution. But the efficiency of ZnCoFe₂O₄ nanoparticles was lower than the ZnCoFe₂O₄@Ch. Because the hydroxyl and amine functional groups present in the ZnCoFe₂O₄@Ch structure help the TC adsorption process. Following the experiments, the significance level between the antibiotic adsorption efficiency with chitosan, ZnCoFe₂O₄, and ZnCoFe₂O₄@Ch was analyzed by ANOVA using the software SPSS v. 21. It was discovered that there was a significant difference ($p < 0.046$) between the adsorption efficiency of these adsorbents.

4. Conclusion

In summary, ZnCoFe₂O₄@Ch as a recyclable and environmentally friendly magnetic nanohybrid adsorbent was prepared with the microwave-assisted co-precipitation method. The structural characteristics of the produced ZnCoFe₂O₄@Ch were investigated by using various analyses. The nano-scale adsorbent was synthesized as homogenous, lump-free, quasi-spherical nanoparticles with large surface areas and magnetic properties while preserving the crystalline structure. Tetracycline was removed from synthetic and real wastewater samples by ZnCoFe₂O₄@Ch in 93 % and 80 %, respectively, under optimal conditions. Tetracycline adsorption is confirmed by adsorption kinetics and isotherms to be compatible with *pseudo*-second-order kinetic and the Freundlich isotherm, respectively. The adsorption process is exothermic, accompanied by the reduction of entropy and spontaneously, according to the findings of the analysis of thermodynamic data. After five cycles of reuse and recovery, the synthesized magnetic nanocomposite was able to remove 65 % of TC from the aqueous environment. Tetracycline may be eliminated from contaminated water and wastewater from the pharmaceutical industries using this magnetic nanocomposite. The findings of the study showed that various polysaccharides, including methylcellulose, carboxymethyl cellulose, and starch, may one day be utilized to change various magnetic nano adsorbents and be incorporated into the development of useful magnetic nanocomposites. These polysaccharides may be utilized to modify ferritic metal spinels, which can then be employed as heterogeneous magnetic nanocatalysts or magnetic nano adsorbents to remove diverse organic and inorganic contaminants from contaminated water environments. Also, the limitation statements include the following things that need more research in this field.

- The operating parameters of the batch adsorption, could handle a larger volume of wastewater?
- Would the operating parameters be feasible for real applications?
- Could the adsorbent effectively remove the pollutants if the concentration range differs?
- Is doing this process economically justified?

Data availability statement

Data will be made available on request.

Declaration of interest's statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional information

No additional information is available for this paper.

Approval of the submitted version of the manuscript

Please check this box to confirm that all co-authors have read and approved the version of the manuscript that is submitted. Signatures are not required.

CRediT authorship contribution statement

Alireza Nasiri: Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Najmeh Golestani:** Writing – original draft, Validation, Software, Methodology, Data curation. **Saeed Rajabi:** Writing – review & editing, Formal analysis, Data curation. **Majid Hashemi:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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